

Etienne Gauthier

 GitHub |  LinkedIn |  Website |  etienne.gauthier@inria.fr

SUMMARY

I am a second-year PhD student in the Sierra team at Inria and ENS Ulm, under the supervision of Francis Bach and Michael I. Jordan. My research focuses on decision-making in multi-agent environments, an area that lies at the intersection of machine learning, statistics, and game theory.

EDUCATION

2024 - 2027 PhD (Computer Science) at **Inria & ENS Ulm**
2023 - 2024 Master's Degree (Mathematics, Vision, Learning) at **ENS Paris-Saclay**
2020 - 2024 Cycle Ingénieur at **École polytechnique**
2018 - 2020 Classes Préparatoires aux Grandes Écoles at **Lycée privé Sainte-Geneviève**

WORK EXPERIENCE

Research Intern, Kyoto University Institute for Advanced Study Mar 2023 – Aug 2023
Kyoto, Japan

- Conducted research under the supervision of Prof. Yasuaki Hiraoka.
- Designed a new kernel for commutative ladders in multidimensional topological data analysis.
- Conducted research in theory of representations of associative algebras.

Annotator, Meta FAIR Aug 2022 – Mar 2023
Paris, France

- Created and validated Lean annotations of mathematical problems from the International Mathematical Olympiad for use in neural theorem proving.
- Supported dataset development for training and evaluating ML models on formal reasoning tasks.

Quantitative Research Intern, Fintica Jun 2022 – Sep 2022
Tel Aviv, Israel

- Developed a Monte Carlo algorithm to identify optimal rolling window sizes, balancing model accuracy and z-score/return similarity.
- Designed a machine learning-based equity investment strategy that outperformed the original approach on key performance indicators.
- Integrated insights from statistical analysis to locate and address suboptimal model performance areas.

Officer Cadet, French Navy Nov 2020 – Apr 2021
Charles de Gaulle Aircraft Carrier

- Completed first-year military training as part of the École polytechnique curriculum.
- Served in an NH90 helicopter unit aboard the *Charles de Gaulle* aircraft carrier.
- Deployed in the Mediterranean Sea, Red Sea, Indian Ocean, and Persian Gulf during the *Clemenceau 21* mission.

TEACHING EXPERIENCE

Teaching Assistant, Université Paris 1 Panthéon-Sorbonne Sep 2022 – Jan 2023

Statistics: Probabilities for 2nd year undergraduate students.

SELECTED HONORS

Google PhD Fellowship in Machine Learning and ML Foundations	Awarded November 2025, \$140,000 funding
ICML 2025 Oral Presentation and Spotlight Paper	Competitive oral session highlighting top papers

PUBLICATIONS

Etienne Gauthier, Francis Bach, and Michael I. Jordan. “Adaptive Coverage Policies in Conformal Prediction”. In: *arXiv preprint arXiv:2510.04318* (2025).

Etienne Gauthier, Francis Bach, and Michael I. Jordan. “Backward Conformal Prediction”. In: *Advances in Neural Information Processing Systems* (2025).

Etienne Gauthier, Francis Bach, and Michael I. Jordan. “E-Values Expand the Scope of Conformal Prediction”. In: *arXiv preprint arXiv:2503.13050* (2025).

Etienne Gauthier, Francis Bach, and Michael I. Jordan. “Statistical Collusion by Collectives on Learning Platforms”. In: *International Conference on Machine Learning* (2025). **Accepted for Talk (top 1%)**.

Hideto Asashiba, Etienne Gauthier, and Enhao Liu. “Interval Replacements of Persistence Modules”. In: *arXiv preprint arXiv:2403.08308* (2024).

TALKS & PRESENTATIONS

- Talk: **Adaptive Coverage Policies in Conformal Prediction**, *Joint Meetings of 2025 Taipei International Statistical Symposium and the 13th ICSA International Conference*, Taipei, Taiwan, Dec 2025.
- Poster: **Backward Conformal Prediction**, *Advances in Neural Information Processing Systems*, San Diego, USA, Dec 2025.
- Talk: **Recent Advances in Conformal Prediction with E-Values**, *Séminaire Parisien de Statistique de l’Institut Henri Poincaré*, Paris, France, Oct 2025.
- Talk: **Recent Advances in Conformal Prediction with E-Values**, *OCEAN Retreat*, Paris, France, Jul 2025.
- Talk & Poster: **Statistical Collusion by Collectives on Learning Platforms**, *International Conference on Machine Learning*, Vancouver, Canada, Jul 2025.
- Talk: **E-Values Expand the Scope of Conformal Prediction**, *3rd Annual Workshop on Game-Theoretic Statistical Inference*, Chennai, India, Jul 2025.
- Talk: **Statistical Collusion by Collectives on Learning Platforms**, *From Matching to Market: A Tale of Mathematics, Economics and Computer Science*, Cargèse, France, Apr 2025.
- Talk: **Statistical Collusion by Collectives on Learning Platforms**, *Chaire Marchés et Apprentissage*, Paris, France, Mar 2025.

ACADEMIC SERVICE

Reviewer: AISTATS 2026